

```
Grammar4
1 grammar Grammar;
2
3 t : e;
4
5 e : e '+' t # Add
6   | e '-' t # Sub
7   | t # Et
8
9 t : t '*' f # Product
10  | t '/' f # Div
11  | f # Tf
12
13 f : '(' e ')' # Paren
14   | DIGIT # Int
15
16 DIGIT : [0-9]+;
17 WS : [ \t\r\n]+ -> skip;
```

```
Interpreter.php
1 use Context\ProductContext;
2 use Context\ParenContext;
3 use Context\IntContext;
4
5 class Interpreter extends GrammarBaseVisitor
6 {
7     public function visitAdd(AddContext $ctx)
8     {
9         $left = $this->visit($ctx->e());
10        $right = $this->visit($ctx->t());
11        return $left + $right;
12    }
13
14    public function visitSub(SubContext $ctx)
15    {
16        $left = $this->visit($ctx->e());
17        $right = $this->visit($ctx->t());
18        return $left - $right;
19    }
20
21    public function visitProduct(ProductContext $ctx)
22    {
23        $left = $this->visit($ctx->t());
24        $right = $this->visit($ctx->f());
25        return $left * $right;
26    }
27
28    public function visitDiv(DivContext $ctx)
29    {
30        $left = $this->visit($ctx->t());
31        $right = $this->visit($ctx->f());
32        return $left / $right;
33    }
34
35    public function visitParen(ParenContext $ctx)
36    {
37        return $this->visit($ctx->e());
38    }
39
40    public function visitInt(IntContext $ctx)
41    {
42        return intval($ctx->DIGIT()->getText());
43    }
44 }
45
```

```
index.php
1 <?php
2
3 require __DIR__ . '/vendor/autoload.php';
4
5 require_once __DIR__ . '/GrammarLexer.php';
6 require_once __DIR__ . '/GrammarParser.php';
7 require_once __DIR__ . '/GrammarVisitor.php';
8 require_once __DIR__ . '/Interpreter.php';
9
10 use Antlr\Antlr4Runtime\InputStream;
11 use Antlr\Antlr4Runtime\CommonTokenStream;
12
13 $input = InputStream::fromString('10 - 2 * 3 + 8 / 4');
14 $lexer = new GrammarLexer($input);
15 $tokens = new CommonTokenStream($lexer);
16 $parser = new GrammarParser($tokens);
17
18 $tree = $parser->tree();
19
20 $interpreter = new Interpreter();
21 $result = $interpreter->visit($tree);
22
23 echo "Result: " . $result . PHP_EOL;
```

10 - 2 * 3 + 8 / 4
resultado esperado: 6